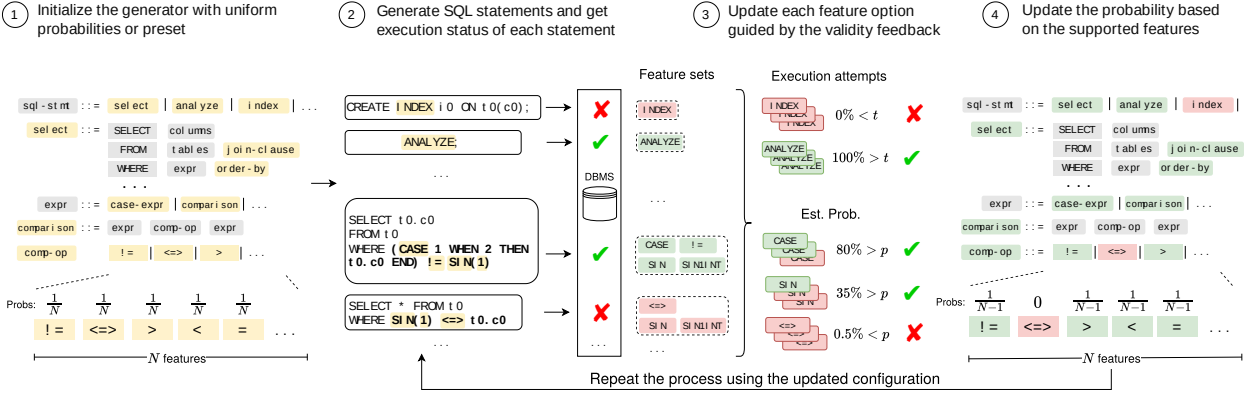

1 Initialize the generator with uniform probabilities or preset
2 Generate SQL statements and get execution status of each statement
3 Update each feature option guided by the validity feedback
4 Update the probability based on the supported features
sql-stmt ::= select | analyze | index | ...
select ::= SELECT columns FROM tables join-clause WHERE expr order-by ...
expr ::= case-expr | comparison | ...
comparison ::= expr comp-op expr
comp-op ::= != | <=> | > | ...
Probs: 1/N 1/N 1/N 1/N 1/N
!= <=> > < = ...
N features
CREATE INDEX i0 ON t0(c0);
ANALYZE;
SELECT t0.c0 FROM t0 WHERE (CASE 1 WHEN 2 THEN t0.c0 END) != SIN(1)
SELECT * FROM t0 WHERE SIN(1) <=> t0.c0
DBMS
Feature sets
INDEX
ANALYZE
CASE != SIN SIN(1,INT
<=> SIN SIN(1,INT
Execution attempts
0% < t
100% > t
Est. Prob.
80% > p
35% > p
0.5% < p
Probs: 1/N-1 0 1/N-1 1/N-1 1/N-1 1/N-1
Repeat the process using the updated configuration